A Nonasymptotic Theory of Gain-Dependent Error Dynamics in Behavior Cloning

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***Abstract*— Behavior cloning (BC) policies on position-controlled robots inherit the closed-loop response of the underlying PD controller, yet the nonasymptotic finite-horizon consequences of controller gains for BC failure remain open. We show that independent sub-Gaussian action errors propagate through the gain-dependent closed-loop dynamics to yield sub-Gaussian position errors whose proxy matrix $X_\infty(K)$ governs the failure tail. The probability of horizon-T task failure factorizes into a gain-dependent amplification index $\Gamma_T(K)$ and the validation loss plus a generalization slack, so training loss alone cannot predict closed-loop performance. Under shape-preserving *upper-bound* structural assumptions the proxy admits the scalar bound $X_\infty(K) \preceq \Psi(K)\bar{X}$ with $\Psi(K)$ decomposed into label difficulty, injection strength, and contraction, ranking the four canonical regimes with compliant-overdamped (CO) tightest, stiff-underdamped (SU) loosest, and the stiff-overdamped versus compliant-underdamped ordering system-dependent. For the canonical scalar second-order PD system the closed-form continuous-time stationary variance $X_\infty^c(\alpha,\beta) = \sigma^2\alpha/(2\beta)$ is strictly monotone in stiffness and damping over the *entire* stable orthant, covering both underdamped and overdamped regimes, and the exact zero-order-hold (ZOH) discretization inherits this monotonicity. The analysis gives a nonasymptotic finite-horizon extension of the gain-dependent error-attenuation explanation in [1].**

Index Terms. Behavior cloning, controller gains, sub-Gaussian concentration, imitation learning.

I. Introduction

Position-controlled action spaces have become the standard interface for learned manipulation policies [2, 3, 4]. At each step a policy $\hat{\pi}_K$ outputs a position target $a_t = \hat{\pi}_K(s_t) \in \mathbb{R}^n$ that a PD controller [5] converts to motor torques via

$$\tau_t = K_p(a_t - q_t) - K_d\,\dot{q}_t + g(q_t), \tag{1}$$

where $q_t, \dot{q}_t \in \mathbb{R}^n$ are the measured joint positions and velocities, $K_p, K_d \in \mathbb{R}^{n\times n}$ are the diagonal positive-definite proportional and derivative gain matrices, and $g(q_t) \in \mathbb{R}^n$ is gravity compensation. Despite the ubiquity of this interface, the choice of gains (K_p, K_d) for *learning* pipelines remains largely heuristic.

Recent work by Bronars et al. [1] systematically tunes controller gains across BC, reinforcement learning, and sim-to-real transfer. The salient finding is that BC benefits from *compliant* (low K_p) and *overdamped* (high K_d) gains. Policies trained under these settings achieve higher closed-loop success rates despite exhibiting *higher* validation loss, a contrast that Bronars et al. explain through the error-attenuation properties of the closed-loop PD controller and a derived stationary variance. Their open-loop noise-injection experiments isolate this mechanism empirically. What remains open is a nonasymptotic finite-horizon theory that connects the same mechanism to validation loss, generalization slack, and failure probabilities. Parallel theoretical work identifies fundamental challenges in continuous-action imitation learning, including sensitivity to function approximation error [6] and noise-driven amplification during autoregressive rollout [7], but these lines do not model the controller through which actions are physically executed.

The classical compounding-error analysis pioneered by Ross and Bagnell [8] and extended in DAgger [9] bounds the policy-versus-expert performance gap by the horizon and the per-step error, treating the dynamics as a black box. Spencer et al. [10] characterize three regimes of covariate shift, again without reference to the gain-dependent structure of the closed loop. In control theory, the propagation of stochastic disturbances through linear systems is well understood via Lyapunov equations and $\mathcal{H}_2/\mathcal{H}_\infty$ norms [11], and recent work on learning-based control [12, 13] connects sample complexity and regret to system-theoretic quantities. These analyses target reinforcement learning or adaptive control, however, not the offline BC pipeline where a fixed policy is deployed open-loop with respect to the expert.

We bridge these two lines of research. The key distinction from classical compounding-error analysis [8, 9] is that our bounds depend explicitly on the controller gains through the closed-loop matrices A_K, B_K, and reveal that the relevant quantity for closed-loop performance is not the prediction loss alone but its product with a gain-dependent amplification.

Our contributions are fourfold as follows:

1) **Sub-Gaussian propagation** (Theorem 1). Independent sub-Gaussian action errors yield a sub-Gaussian position error whose proxy matrix $X_t(K)$ is the gain-dependent Lyapunov sum.
2) **Failure bound via validation loss** (Theorem 3). The probability of horizon-T task failure factorizes into a gain-dependent amplification index $\Gamma_T(K)$ and the validation loss plus generalization slack, showing that higher validation loss can coexist with a lower failure bound.
3) **Scalar ordering and regime comparison** (Theorems 5 and 7). Under shape-preserving *upper-bound* assumptions, the proxy obeys $X_\infty(K) \preceq \Psi(K)\bar{X}$ with $\Psi(K) = b(K)l(K)/(1-\rho_*(K)^2)$, and monotonicity of Ψ ranks the four canonical regimes.
4) **Global canonical monotonicity** (Theorem 9). For the

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canonical scalar second-order PD system, $X^{\mathrm{c}}_\infty(\alpha,\beta) = \sigma^2\alpha/(2\beta)$ is strictly monotone in stiffness and damping over the *entire* stable orthant, and the exact ZOH discretization inherits this monotonicity.

II. Related Works

a) Behavior cloning theory: Behavior cloning [14, 15] learns a policy by supervised regression on expert state-action pairs. The foundational analysis of Ross and Bagnell [8] shows that the expected cost of the learned policy grows quadratically with horizon T from compounding errors, motivating interactive approaches such as DAgger [9]. Subsequent work refines these bounds under structural assumptions [10, 16], sharpens the horizon dependence [17], and links low-level dynamical stability to task-level guarantees [18]. Building on the controller-level mechanism identified in [1], our analysis treats the controller gain as a structural variable in nonasymptotic BC error propagation.

b) Impedance control: The interaction between compliance and task performance has a long history in robotics [19, 20, 21], with variable impedance learning [22, 23, 24] adapting compliance during execution. Our contribution is orthogonal. We analyze how *fixed* gains modulate the propagation of prediction errors during BC rollout.

c) Controller design for robot learning: The choice of action space and controller gains has long been recognized as a critical design axis for robot learning [25, 26, 27]. Kim et al. [28] show that torque-level control improves task-agnostic transfer, and Bronars et al. [1] provide a systematic study of how PD gains affect BC, RL, and sim-to-real transfer, identifying compliant-overdamped settings as the BC optimum. They also propose and test a mechanistic explanation based on closed-loop error attenuation, using open-loop noise-injection experiments to isolate the effect and deriving the stationary variance that captures it. The theoretical analysis below extends that account to sub-Gaussian propagation, finite-horizon failure bounds, and generalization slack.

III. Problem Formulation

A. Gain-Dependent Closed-Loop Error Dynamics

Consider a robot with n joints and inertia $M \in \mathbb{R}^{n\times n}$ operating under the PD controller (1) with gain setting $K = (K_p, K_d)$. Let $q_t^\star$ denote the expert joint trajectory and $a_t^\star$ the expert action. Define the position error $e_t := q_t - q_t^\star$ and the prediction error $\xi_t := \hat{\pi}_K(s_t) - a_t^\star$. Substituting $a_t = a_t^\star + \xi_t$ into (1) and linearizing about the expert trajectory yields

$$M\,\ddot{e}_t = -K_p\,e_t - K_d\,\dot{e}_t + K_p\,\xi_t, \tag{2}$$

so the action error enters scaled by K_p and stiffer gains amplify both the restoring force on e_t and the injection of action errors. Setting $x_t := [e_t^\top, \dot{e}_t^\top]^\top \in \mathbb{R}^{2n}$ gives the continuous state-space form

$$\dot{x}_t = A_K^{\mathrm{c}} x_t + B_K^{\mathrm{c}}\xi_t, \qquad e_t = Cx_t, \tag{3}$$

with $A_K^{\mathrm{c}} = \begin{bmatrix} 0 & I_n \\ -M^{-1}K_p & -M^{-1}K_d \end{bmatrix}$, $B_K^{\mathrm{c}} = \begin{bmatrix} 0 \\ M^{-1}K_p \end{bmatrix}$, and $C = [I_n, 0]$.

The policy emits a new target every Δt seconds and the controller holds it constant between updates, so the sampled state obeys

$$x_{t+1} = A_K x_t + B_K\xi_t, \qquad e_t = Cx_t, \tag{4}$$

with matrices given by the *exact zero-order-hold (ZOH)* sampling of (3),

$$A_K = e^{A_K^{\mathrm{c}}\Delta t}, \quad B_K = \left(\int_0^{\Delta t} e^{A_K^{\mathrm{c}} s}\,ds\right) B_K^{\mathrm{c}}. \tag{5}$$

We consider gains for which the sampled closed loop is asymptotically stable, $\rho(A_K) < 1$.

B. Action Prediction Error Model

We model the prediction errors $\{\xi_t\}$ as *independent, mean-zero, sub-Gaussian* random vectors. There exists a PSD matrix $\Sigma_K^{\mathrm{roll}} \succeq 0$ such that

$$\mathbb{E}\big[\exp(\lambda^\top\xi_t)\big] \le \exp\big(\tfrac{1}{2}\lambda^\top\Sigma_K^{\mathrm{roll}}\lambda\big), \quad \forall\,\lambda \in \mathbb{R}^m. \tag{6}$$

The proxy covariance Σ_K^{roll} governs the tail behavior, with the subscript K reflecting that different gains induce different action labels and hence different residual statistics. Independence is justified in the early-rollout regime before significant covariate shift, and sub-Gaussianity accommodates bounded errors and light-tailed neural network residuals. The *rollout-local population action MSE* is $L_{\mathrm{roll}}(K) := \mathbb{E}[\|\xi_t\|_2^2] = \mathrm{tr}(\Sigma_K^{\mathrm{roll}})$.

C. Task Success, Validation Loss, and Amplification

A rollout is successful if the position error stays within a success tube of radius $r > 0$ over horizon T. The complementary failure event is

$$\mathrm{Fail}_T := \{\exists\, t \le T : \|e_t\|_2 \ge r\}. \tag{7}$$

After training, the empirical validation loss over N_{va} held-out pairs $\{(\tilde{s}_j, \tilde{a}_j)\}$ is $\hat{L}_{\mathrm{va}}(K) := N_{\mathrm{va}}^{-1}\sum_j \|\hat{\pi}_K(\tilde{s}_j) - \tilde{a}_j\|_2^2$, and the training loss $\hat{L}_{\mathrm{tr}}(K)$ is defined analogously. A standard generalization argument gives, with probability at least $1-\delta$,

$$L_{\mathrm{roll}}(K) \le \hat{L}_{\mathrm{va}}(K) + \varepsilon_{\mathrm{gen}}(K,\delta). \tag{8}$$

We define the *finite-horizon amplification index*

$$\Gamma_T(K) := \max_{0\le t\le T}\sum_{s=0}^{t-1}\|CA_K^s B_K\|_{\mathrm{op}}^2, \tag{9}$$

which measures the worst-case cumulative operator-norm gain from action errors to position errors and depends on the gains through A_K and B_K.

IV. Main Results

We present four results in a logical chain: a sub-Gaussian propagation theorem, a closed-loop failure bound, a scalar ordering reduction in the upper-bound direction, and a regime comparison.

A. Sub-Gaussian Error Propagation

Theorem 1 lifts the per-step sub-Gaussian assumption on the action error to a horizon-wide statement about the position error. The lift is a direct consequence of linearity: a weighted sum of independent sub-Gaussians is sub-Gaussian, and the weights are exactly the closed-loop impulse response at lag s. The Lyapunov limit $S_\infty(K)$ is the standard quantity for disturbance-to-state propagation in discrete LTI systems, and the role of this paper is to track how every ingredient depends on the controller gain, so tuning (K_p, K_d) reshapes the entire tail of e_t.

Theorem 1 (Sub-Gaussian propagation)**.** *Consider* (4) *with $x_0 = 0$ and action errors satisfying* (6)*. Define the* proxy matrix

$$S_t(K) := \sum_{s=0}^{t-1} A_K^s B_K \Sigma_K^{\mathrm{roll}} B_K^\top (A_K^\top)^s, \tag{10}$$

and its position-error projection $X_t(K) := C\, S_t(K)\, C^\top$. Then e_t is mean-zero sub-Gaussian with proxy $X_t(K)$. For all $u \in \mathbb{R}^n$,

$$\mathbb{E}\big[\exp(u^\top e_t)\big] \le \exp\big(\tfrac{1}{2} u^\top X_t(K)\, u\big)\,, \tag{11}$$

$$\mathbb{P}\big(|u^\top e_t| \ge r\big) \le 2 \exp\left(-\frac{r^2}{2\, u^\top X_t(K)\, u}\right). \tag{12}$$

If $\rho(A_K) < 1$, then $S_t(K) \to S_\infty(K)$, the unique PSD solution of the discrete Lyapunov equation

$$S_\infty(K) = A_K S_\infty(K) A_K^\top + B_K \Sigma_K^{\mathrm{roll}} B_K^\top, \tag{13}$$

and the stationary proxy is $X_\infty(K) = C\, S_\infty(K)\, C^\top$.

Proof. Unrolling (4) from $x_0 = 0$ gives $e_t = \sum_{s=0}^{t-1} C A_K^s B_K \xi_{t-1-s}$. For $u \in \mathbb{R}^n$ and $h_s := B_K^\top (A_K^\top)^s C^\top u$, we have $u^\top e_t = \sum_s h_s^\top \xi_{t-1-s}$. By independence and the sub-Gaussian MGF bound,

$$\begin{aligned}\mathbb{E}\left[e^{u^\top e_t}\right] &\le \prod_s \exp\big(\tfrac{1}{2} h_s^\top \Sigma_K^{\mathrm{roll}} h_s\big) \\ &= \exp\big(\tfrac{1}{2} u^\top X_t(K) u\big)\,.\end{aligned} \tag{14}$$

The tail bound follows from the Chernoff method, and Schur stability with the discrete Lyapunov theorem yields convergence to $X_\infty(K)$. □

Corollary 2 (Euclidean tail bound)**.** *Under Theorem 1,*

$$\mathbb{P}(\|e_t\|_2 \ge r) \le 2n \exp\left(-\frac{r^2}{2n\, \lambda_{\max}(X_t(K))}\right). \tag{15}$$

Proof. Diagonalize $X_t = Q \Lambda Q^\top$ and set $y = Q^\top e_t$. If $\|y\|_2 \ge r$ then some $|y_i| \ge r/\sqrt{n}$, and each y_i is sub-Gaussian with proxy $\lambda_i \le \lambda_{\max}(X_t)$. A union bound over coordinates gives the result. □

The dimensional factor n enters twice, once as a linear prefactor and once inside the denominator of the exponent. In the scalar case $n = 1$ both factors disappear and the bound reduces to the one-dimensional Chernoff tail. For robot manipulators n is typically between six and twelve, and the corresponding slack is mild on a logarithmic scale.

B. Closed-Loop Failure Bound

Theorem 3 (Failure bound via validation loss)**.** *Under the framework of Section III, with probability at least $1-\delta$ over training randomness,*

$$\begin{aligned}&\mathbb{P}(\mathrm{Fail}_T \mid \hat{\pi}_K) \le \\ &2n(T{+}1) \exp\left(-\frac{r^2}{2n\, \Gamma_T(K)\Big(\hat{L}_{\mathrm{va}}(K) + \varepsilon_{\mathrm{gen}}(K,\delta)\Big)}\right).\end{aligned} \tag{16}$$

Proof. From Theorem 1, $\lambda_{\max}(X_t(K)) \le \sum_s \|C A_K^s B_K\|_{\mathrm{op}}^2\, \lambda_{\max}(\Sigma_K^{\mathrm{roll}}) \le \Gamma_T(K)\, L_{\mathrm{roll}}(K)$, where the second inequality uses $\lambda_{\max}(\Sigma_K^{\mathrm{roll}}) \le \mathrm{tr}(\Sigma_K^{\mathrm{roll}}) = L_{\mathrm{roll}}(K)$. The generalization bound (8) replaces $L_{\mathrm{roll}}(K)$ by $\hat{L}_{\mathrm{va}}(K) + \varepsilon_{\mathrm{gen}}(K,\delta)$ with probability $1-\delta$. Substituting into (15) and union-bounding over $t = 0, 1, \ldots, T$ yields (16). □

The exponent of (16) is governed by the product $\Gamma_T(K)\,(\hat{L}_{\mathrm{va}}(K) + \varepsilon_{\mathrm{gen}}(K,\delta))$ rather than by the loss alone, so a gain K_1 with lower training or validation loss than K_2 can incur a strictly worse failure bound once its amplification is correspondingly larger. This matches the attenuation-difficulty tradeoff of [1], in which compliant-overdamped gains attain higher closed-loop success despite higher validation MSE because their amplification is small enough to dominate the product. Appendix A of [1] derives the stationary variance $\sigma^2 K_p/(2K_d)$, and its *State-Space Impact of Policy Errors* corollary uses $\sqrt{K_p/(2K_d)}\,\epsilon_\pi(K)$ to explain the same mechanism; Theorem 3 extends that analysis to a nonasymptotic finite-horizon tail bound carrying the explicit generalization slack $\varepsilon_{\mathrm{gen}}(K,\delta)$.

C. Scalar Ordering Reduction

To compare gain regimes through a single scalar, we impose structural conditions on the error dynamics that yield an *upper bound* on the proxy matrix. Every inequality below is an upper bound, and reversing any one invalidates the chain.

Definition 4 (Shape-preserving upper-bound structure)**.** *The error dynamics* (4) *have* shape-preserving upper-bound structure *if there exist fixed PSD matrices $\bar{W}, \bar{\Sigma} \succeq 0$ and scalar functions $l(K), b(K) > 0$, $\rho_*(K) \in [0,1)$ such that*

(i) label difficulty: $\Sigma_K^{\mathrm{roll}} \preceq l(K)\, \bar{\Sigma}$,

(ii) injection: $B_K \bar{\Sigma} B_K^\top \preceq b(K)\, \bar{W}$,

(iii) Lyapunov contraction: $A_K \bar{W} A_K^\top \preceq \rho_*(K)^2\, \bar{W}$.

Condition (i) bounds the rollout-error proxy at gain K, condition (ii) bounds how strongly action errors are injected into the state space, and condition (iii) certifies $\bar{W}$ as a Lyapunov function for A_K with rate $\rho_*(K)^2$.

Theorem 5 (Scalar ordering reduction)**.** *Under Definition 4, define the* ordering index

$$\Psi(K) := \frac{b(K)\, l(K)}{1 - \rho_*(K)^2}. \tag{17}$$

Then

$$X_\infty(K) \preceq \Psi(K)\,\bar{X}, \qquad \bar{X} := C\,\bar{W}\,C^\top. \tag{18}$$

Consequently, $\lambda_{\max}(X_\infty(K)) \leq \Psi(K)\,\lambda_{\max}(\bar{X})$. If $\Psi(K_1) \leq \Psi(K_2)$, then K_1's proxy upper bound is no larger than K_2's, and substituting into (16) *yields a failure-probability upper bound for K_1 that is no larger than the corresponding upper bound for K_2.*

Proof. By (i), left- and right-multiplying $\Sigma_K^{\mathrm{roll}} \preceq l(K)\bar{\Sigma}$ by B_K and its transpose and applying (ii) gives $B_K\Sigma_K^{\mathrm{roll}}B_K^\top \preceq l(K)b(K)\,\bar{W}$. The map $X \mapsto A_K^s X (A_K^\top)^s$ preserves $\preceq$ for $s \geq 0$, so $A_K^s B_K \Sigma_K^{\mathrm{roll}} B_K^\top (A_K^\top)^s \preceq l(K)b(K)A_K^s\bar{W}(A_K^\top)^s$. Iterating (iii) yields $A_K^s\bar{W}(A_K^\top)^s \preceq \rho_*(K)^{2s}\bar{W}$, and the geometric sum gives $S_\infty(K) \preceq l(K)b(K)\sum_{s\geq 0}\rho_*(K)^{2s}\bar{W} = \Psi(K)\bar{W}$. Projecting through C yields (18), and operator monotonicity of $\lambda_{\max}$ followed by substitution into (16) gives the failure-bound monotonicity. □

The index $\Psi(K)$ assembles three independent penalties: a large label difficulty $l(K)$, a large injection $b(K)$, and slow contraction (large $\rho_*(K)$). A gain can remain optimal even when its label difficulty is the highest of the four candidates, provided the injection and contraction terms are correspondingly smaller. This three-way competition is precisely what makes the BC gain-tuning problem nontrivial, and Theorem 5 reduces the resolution of that competition to a single scalar comparison. Importantly, the three penalties are not symmetric in the gains: the injection $b(K)$ is typically a strictly increasing function of K_p (stiffer gains inject action errors more aggressively), while the contraction $\rho_*(K)$ is typically a strictly decreasing function of K_d (stronger damping tightens the Lyapunov rate). The label-difficulty term $l(K)$ is the only component with a qualitatively data-driven behavior: harder labels are induced by compliant settings because the policy must compensate for richer closed-loop trajectories.

Corollary 6 (Finite-horizon convergence)**.** *Under Definition 4, the truncation error of the proxy admits the upper bound*

$$X_\infty(K) - X_t(K) \preceq \frac{\rho_*(K)^{2t}}{1-\rho_*(K)^2}\,b(K)\,l(K)\,\bar{X}, \tag{19}$$

and consequently $\|X_\infty(K) - X_t(K)\|_{\mathrm{op}} \leq \rho_*(K)^{2t}\,\Psi(K)\,\|\bar{X}\|_{\mathrm{op}}$, so the approximation error decays geometrically at rate $\rho_*(K)^2$ per step.

Proof. $S_\infty(K) - S_t(K) = \sum_{s=t}^\infty A_K^s W_K (A_K^\top)^s$ with $W_K = B_K\Sigma_K^{\mathrm{roll}}B_K^\top$. Repeating the upper-bound chain from the proof of Theorem 5 on the tail sum gives $X_\infty(K) - X_t(K) \preceq l(K)b(K)\rho_*(K)^{2t}/(1-\rho_*(K)^2)\bar{X}$. The operator-norm bound follows since $\|\cdot\|_{\mathrm{op}}$ is monotone under $\preceq$ for PSD matrices. □

D. Regime Ordering

We parameterize gains as $K = (\alpha, \beta)$ where α denotes stiffness (K_p scale) and β denotes damping (K_d scale), and define four canonical regimes

$$\begin{aligned} \mathrm{CO} &= (\alpha_L, \beta_H), & \mathrm{SO} &= (\alpha_H, \beta_H), \\ \mathrm{CU} &= (\alpha_L, \beta_L), & \mathrm{SU} &= (\alpha_H, \beta_L), \end{aligned} \tag{20}$$

with $\alpha_H > \alpha_L$ and $\beta_H > \beta_L$.

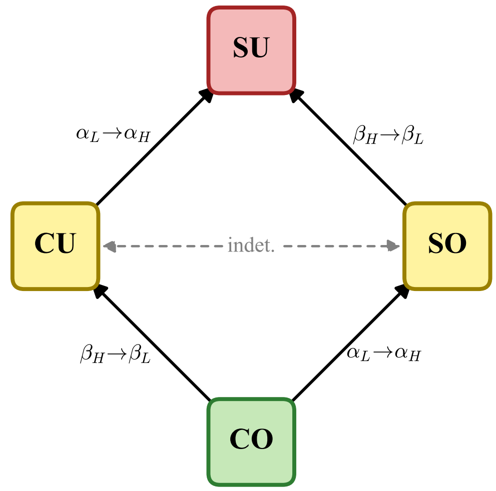

Fig. 1. Hasse diagram of the regime ordering established by Theorem 7. Solid arrows indicate $\Psi(K_1) \leq \Psi(K_2)$. CO is the unique minimum and SU the unique maximum. The dashed line between SO and CU indicates that their ordering is system-dependent.

Theorem 7 (Regime ordering)**.** *Suppose $\Psi(\alpha,\beta)$ is non-decreasing in α and non-increasing in β. Then*

$$\Psi(\mathrm{CO}) \leq \min\{\Psi(\mathrm{SO}),\, \Psi(\mathrm{CU})\}, \tag{21}$$
$$\Psi(\mathrm{SU}) \geq \max\{\Psi(\mathrm{SO}),\, \Psi(\mathrm{CU})\}, \tag{22}$$

so CO admits the tightest failure-bound upper bound and SU the loosest. The SO vs. CU comparison reduces to $\Psi(\alpha_H,\beta_H) \lesseqgtr \Psi(\alpha_L,\beta_L)$ and is therefore system-dependent.

Proof. Direct from monotonicity. $\Psi(\alpha_L,\beta_H) \leq \Psi(\alpha_H,\beta_H)$ by stiffness-monotonicity, and $\Psi(\alpha_L,\beta_H) \leq \Psi(\alpha_L,\beta_L)$ by damping-antimonotonicity. Similarly $\Psi(\alpha_H,\beta_H) \leq \Psi(\alpha_H,\beta_L)$ and $\Psi(\alpha_L,\beta_L) \leq \Psi(\alpha_H,\beta_L)$. The SO-CU comparison involves opposing effects and cannot be resolved without additional information. □

E. Verification for Canonical Second-Order Systems

Theorem 7 reduces the regime ordering to monotonicity of $\Psi(K)$. We verify this monotonicity *directly* for the canonical second-order PD system without invoking the shape-preserving abstraction. The continuous-time analysis yields a closed-form expression that is monotone over the entire stable gain space, covering both underdamped and overdamped regimes, and the exact ZOH sampling used in Section V inherits the same monotonicity.

Lemma 8 (Continuous-time stationary covariance)**.** *For the scalar ($n = 1$) continuous-time error dynamics* (3) *with mass m and gains $K_p = \alpha$, $K_d = \beta$ driven by a* *white action-error process of intensity σ^2, the stationary covariance of x_t is the unique PSD solution of $A_K^{\mathrm{c}}P + P(A_K^{\mathrm{c}})^\top + B_K^{\mathrm{c}}\sigma^2(B_K^{\mathrm{c}})^\top = 0$, given by*

$$P^{\mathrm{c}}(\alpha,\beta) = \begin{bmatrix} \dfrac{\sigma^2\alpha}{2\beta} & 0 \\ 0 & \dfrac{\sigma^2\alpha^2}{2\beta\, m} \end{bmatrix}. \tag{23}$$

The position-error stationary variance is therefore

$$X_\infty^{\mathrm{c}}(\alpha,\beta) = CP^{\mathrm{c}}C^\top = \frac{\sigma^2\alpha}{2\beta}. \tag{24}$$

Proof. Write $A_c = \left[\begin{smallmatrix} 0 & 1 \\ -\alpha/m & -\beta/m \end{smallmatrix}\right]$, $B_c = (\alpha/m)e_2$, and $P = \left[\begin{smallmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{smallmatrix}\right]$. Expanding $A_cP + PA_c^\top + B_c\sigma^2B_c^\top = 0$ entry-wise gives the three independent equations $2p_{12} = 0$, $p_{22} = (\alpha/m)p_{11}$, and $-2(\beta/m)p_{22} + (\alpha/m)^2\sigma^2 = 0$, solved in order to obtain $p_{12} = 0$, $p_{22} = \sigma^2\alpha^2/(2\beta m)$, and $p_{11} = \sigma^2\alpha/(2\beta)$. Hurwitz stability of A_c gives uniqueness of the PSD solution. ☐

Theorem 9 (Global monotonicity in canonical systems)**.** *The scalar canonical stationary variance $X_\infty^{\mathrm{c}}(\alpha,\beta) = \sigma^2\alpha/(2\beta)$ is strictly increasing in stiffness α and strictly decreasing in damping β over the entire stable region $\{(\alpha,\beta) \in \mathbb{R}^2 : \alpha,\beta > 0\}$, covering both underdamped ($\beta^2 < 4m\alpha$) and overdamped ($\beta^2 \geq 4m\alpha$) regimes. The canonical regime ordering of Theorem 7 therefore holds with $\Psi^{\mathrm{c}}(\alpha,\beta) := X_\infty^{\mathrm{c}}(\alpha,\beta)/\bar{X}^{\mathrm{c}}$ for any positive scalar reference $\bar{X}^{\mathrm{c}}$.*

Proof. From (24), $\partial_\alpha X_\infty^{\mathrm{c}} = \sigma^2/(2\beta) > 0$ and $\partial_\beta X_\infty^{\mathrm{c}} = -\sigma^2\alpha/(2\beta^2) < 0$ on the positive orthant. The Hurwitz condition for the canonical scalar system is exactly $\alpha,\beta > 0$, so the closed form covers the entire stable region without an underdamped/overdamped split, and the four regimes inherit the ordering of Theorem 7. ☐

Working in continuous time keeps the Lyapunov solution in closed form and avoids any branching on the underdamped/overdamped split, while inheritance through exact ZOH discretization matches the input matrix used throughout Section V. The scalar $\sigma^2K_p/(2K_d)$ is the formal stationary-variance quantity derived in the analytical appendix of [1] under a one-degree-of-freedom Gaussian white-noise model. Lemma 8 recovers the same quantity from the continuous-time Lyapunov equation, and Theorem 9 together with Proposition 10 extends the identity to sub-Gaussian forcing and to strict monotonicity on the entire stable orthant under exact ZOH.

Proposition 10 (Discrete-time inheritance under ZOH)**.** *For the canonical scalar system sampled by* (5) *with control period $\Delta t > 0$, the discrete stationary proxy $X_\infty^{\mathrm{d}}(\alpha,\beta,\Delta t)$ is smooth in (α,β) on the open Schur-stable region, and $X_\infty^{\mathrm{d}}(\alpha,\beta,\Delta t)/\Delta t \to X_\infty^{\mathrm{c}}(\alpha,\beta)$ as $\Delta t \to 0^+$. For control periods small enough that the leading-order continuous-time term dominates, X_∞^{d} inherits the global strict monotonicity of X_∞^{c} in (α,β), covering the gain ranges used at typical BC control rates of* 50–1000 *Hz.*

Proof. The ZOH matrices $A_K(\Delta t) = e^{A_K^{\mathrm{c}}\Delta t}$ and $B_K(\Delta t) = (\int_0^{\Delta t} e^{A_K^{\mathrm{c}}s}ds)B_K^{\mathrm{c}}$ are smooth in $(\alpha,\beta,\Delta t)$ on the stable region, with $A_K = I + A_K^{\mathrm{c}}\Delta t + O(\Delta t^2)$ and $B_K = \Delta t\, B_K^{\mathrm{c}} + O(\Delta t^2)$. Substituting into (13) and expanding to leading order yields $A_K^{\mathrm{c}}S_\infty^{\mathrm{d}} + S_\infty^{\mathrm{d}}(A_K^{\mathrm{c}})^\top + \Delta t\, B_K^{\mathrm{c}}\sigma^2(B_K^{\mathrm{c}})^\top + O(\Delta t^2) = 0$, so $S_\infty^{\mathrm{d}}/\Delta t \to P^{\mathrm{c}}$. Strict inequalities are open under continuous perturbation, so the strict monotonicity of X_∞^{c} from Theorem 9 carries over, and the numerical sweep of Fig. 3 confirms it on the experimental gain range at $\Delta t = 0.02\,\mathrm{s}$. ☐

Corollary 11 (Explicit failure bound, canonical case)**.** *For the canonical scalar second-order system, the failure-bound exponent of Theorem 3 admits the explicit form*

$$\mathbb{P}(\mathrm{Fail}_T) \leq 2(T{+}1)\exp\!\left(-\frac{r^2\,\beta}{\alpha\,\Delta t\, L_{\mathrm{roll}}(K)}\right) + o(\Delta t), \tag{25}$$

where the remainder collects Δt^2 and higher discretization corrections via Proposition 10. The exponent therefore scales as β/α for fixed validation loss, so increasing damping or decreasing stiffness sharpens the bound.

Proof. Proposition 10 gives $X_\infty^{\mathrm{d}}/\Delta t = X_\infty^{\mathrm{c}}(\alpha,\beta) + O(\Delta t)$, and Lemma 8 identifies the leading term as $\sigma^2\alpha/(2\beta)$. Substituting $X_\infty^{\mathrm{d}} = \Delta t\,\sigma^2\alpha/(2\beta) + o(\Delta t)$ into (15) with $n = 1$ and $\sigma^2 = L_{\mathrm{roll}}(K)$ yields (25). ☐

For a fixed prediction-error budget, equation (25) depends only on the ratio β/α. Doubling both gains leaves the bound unchanged, which decouples the gain-tuning question from the absolute scale of the controller.

Corollary 12 (Connection to the squared $\mathcal{H}_2$ norm)**.** *For the canonical second-order system, the closed-form continuous-time stationary variance equals the squared $\mathcal{H}_2$ norm of the disturbance-to-position transfer function $H_K(s) = (\alpha/m)/(s^2 + (\beta/m)s + \alpha/m)$ scaled by σ^2, and the amplification index of Theorem 3 converges in the long-horizon limit to $\lim_{T\to\infty}\Gamma_T(K) = \|\mathcal{G}_K\|_{\mathcal{H}_2}^2$. Minimizing $\Psi(K)$ recovers the classical $\mathcal{H}_2$-optimal gain selection rule as the failure-bound minimizer.*

Proof. The transfer function $H_K(s)$ has natural frequency $\omega_n = \sqrt{\alpha/m}$ and damping ratio $\zeta = \beta/(2\sqrt{m\alpha})$. The standard $\mathcal{H}_2$ formula gives $\|H_K\|_{\mathcal{H}_2}^2 = (\alpha/m)^2/(4\zeta\omega_n^3) = \alpha/(2\beta)$, matching Lemma 8 after the noise scaling. The asymptotic identification of Γ_T is standard for Schur-stable LTI systems [11]. ☐

V. Numerical Illustration

We illustrate the framework on the canonical scalar second-order system, verifying sub-Gaussian propagation, the failure bound, and the regime ordering. The closed-loop matrices are obtained by exact ZOH sampling (5), matching Proposition 10.

A. Setup

Consider unit mass ($m = 1\,\mathrm{kg}$) under PD control sampled at $\Delta t = 0.02\,\mathrm{s}$ (50 Hz). We evaluate the four canonical gain regimes of (20) with $\alpha_L = 50$, $\alpha_H = 100$, $\beta_L = 20$, $\beta_H =$

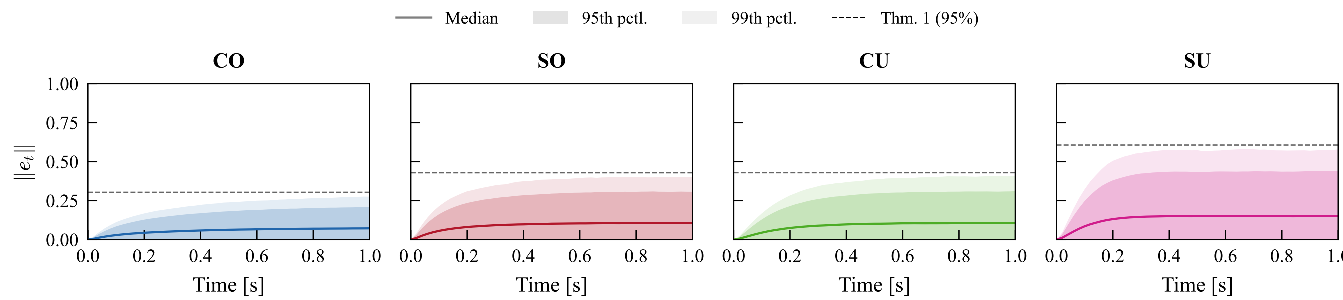

Fig. 2. Monte Carlo position-error envelopes (N = 50,000 rollouts) for the four gain regimes under exact ZOH discretization. Shaded bands show the 95th and 99th percentiles. The dashed line is the steady-state 95th-percentile bound from Theorem 1. All panels share the same vertical scale. CO yields the tightest envelopes, confirming the predicted ordering.

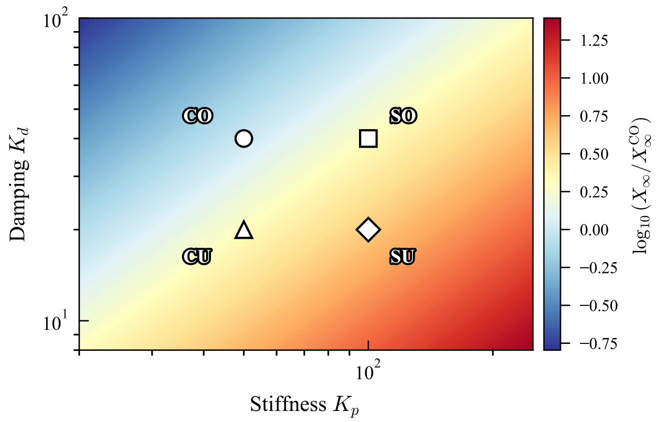

Fig. 3. Stationary proxy $X_\infty(K)$ (normalized by X_∞^{CO}, log scale) over the gain parameter space, computed via exact ZOH discretization at $\Delta t = 0.02\,\mathrm{s}$. Markers locate the four canonical regimes. The landscape is monotone increasing in K_p and decreasing in K_d over the entire stable region, consistent with Theorem 9 and Proposition 10.

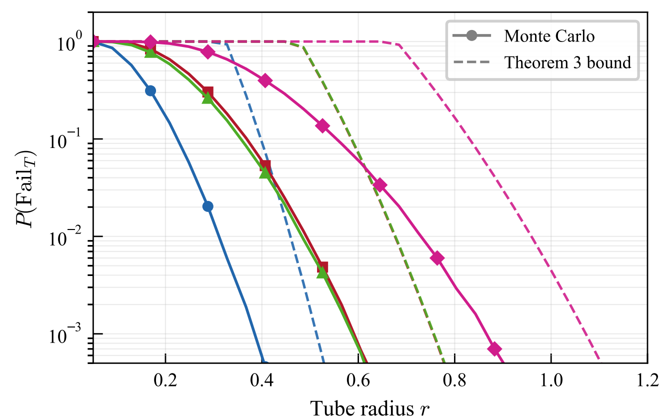

Fig. 4. Empirical failure rate $\hat{P}(\mathrm{Fail}_T)$ (solid) versus the Theorem 3 upper bound (dashed) as a function of the success-tube radius r for the four canonical regimes. The bound dominates the Monte Carlo curve in every regime, and the regime ordering CO $\prec$ SO $\approx$ CU $\prec$ SU is preserved at all r.

TABLE I

SYSTEM QUANTITIES AND EMPIRICAL FAILURE RATES FOR THE SCALAR CANONICAL SYSTEM ($m = 1\,\mathrm{KG}$, $\Delta t = 0.02\,\mathrm{S}$, $T = 50$, $r = 0.3$). X_∞^{c} FROM (24), X_∞^{d} FROM THE ZOH DISCRETE LYAPUNOV.

	CO	**SO**	**CU**	**SU**
K_p	50	100	50	100
K_d	40	40	20	20
$\rho(A_K)$	0.974	0.948	0.943	0.819
$X_\infty^{\mathrm{c}} = \sigma^2 K_p/(2K_d)$	0.625	1.250	1.250	2.500
X_∞^{d} (ZOH discrete)	0.012	0.025	0.025	0.050
$X_\infty^{\mathrm{d}}/X_\infty^{\mathrm{d,CO}}$	**1.0**	2.0	2.0	**4.0**
$\hat{P}(\mathrm{Fail}_T)$	0.01	0.26	0.22	0.75

40. Action prediction errors are drawn i.i.d. from $\mathcal{N}(0, 1)$, satisfying (6) with $\Sigma_K^{\mathrm{roll}} = 1$. For each regime we simulate $N = 50{,}000$ independent rollouts of $T = 50$ steps ($1\,\mathrm{s}$ task horizon), and compute the stationary proxy $X_\infty(K)$ from the discrete Lyapunov equation (13).

B. Sub-Gaussian Error Propagation

Fig. 2 shows the position-error magnitude $\|e_t\|$ across regimes, with median, 95th-percentile, and 99th-percentile envelopes from the Monte Carlo ensemble. The dashed line is the steady-state 95th-percentile threshold predicted by Theorem 1 via $r_{95} = \sqrt{2X_\infty(K)\ln 40}$, derived from (12) with $n = 1$. The empirical 95th percentile stays inside the predicted bound in every regime, confirming sub-Gaussian propagation. The error magnitudes differ systematically: CO yields steady-state errors roughly half those of SO or CU and four times smaller than SU, despite identical action-error distributions across the four regimes.

C. Failure Bound and Regime Ordering

Table I reports both the closed-form continuous-time stationary variance $X_\infty^{\mathrm{c}} = \sigma^2\alpha/(2\beta)$ and the exact discrete ZOH proxy X_∞^{d}. Both quantities respect the predicted ordering CO $\prec$ SO $\approx$ CU $\prec$ SU, with the near-equality of SO and CU exemplifying the system-dependent SO/CU comparison. The Monte Carlo failure rates corroborate the ordering: at $r = 0.3$, CO fails on 1% of rollouts, SO and CU on 26% and 22%, and SU on 75%. Notably CO has a *larger* spectral radius (0.974) than SU (0.819) yet achieves a fourfold smaller proxy because its injection is proportionally weaker, reflecting the three-way competition between label difficulty, injection strength, and contraction inside $\Psi(K)$.

Fig. 3 sweeps $X_\infty(K)$ over the (K_p, K_d) parameter space. The landscape is monotone increasing in stiffness and decreasing in damping over the *entire* sampled region, covering both underdamped and overdamped regimes, with CO at the global minimum and SU at the maximum, consistent with Theorem 9.

D. Test of the Failure Bound

Fig. 4 compares the empirical failure rate $\hat{P}(\mathrm{Fail}_T)$ with the upper bound of Theorem 3 as the success-tube radius r varies. For each regime, the dashed curve plots $\min\{1,\, 2(T{+}1)\exp(-r^2/(2\Gamma_T(K)L_{\mathrm{roll}}(K)))\}$ from Corollary 2 with $L_{\mathrm{roll}} = 1$ and the empirical $\Gamma_T(K)$ from (9). The bound dominates the Monte Carlo curve uniformly across regimes and tube radii, with CO retaining the largest gap (well-conditioned tail) and SU sitting closest to its bound, and the relative ordering matches Theorem 7 at every r. The failure-bound exponent is therefore qualitatively correct as a comparator and quantitatively conservative for absolute prediction. The sub-Gaussian assumption is the dominant source of looseness, since L_{roll} replaces the per-coordinate proxy by its trace and the union over $T+1$ steps adds an unconditional log-factor.

VI. Discussion

A. Gain-Selection Criterion

Theorem 3 turns the attenuation-difficulty tradeoff of [1] into a finite-horizon decision rule. Rather than minimizing the training or validation loss, practitioners should minimize the product $\Gamma_T(K) \cdot (\hat{L}_{\mathrm{va}}(K) + \varepsilon_{\mathrm{gen}}(K, \delta))$, equivalently $\Psi(K)$ when shape-preserving structure holds or $X^{\mathrm{c}}_{\infty}(K)$ for canonical second-order systems. Theorems 7 and 9 recommend the compliant-overdamped regime in agreement with [1]. In our bound, the higher-validation-MSE yet better-closed-loop-performance pattern appears inside the product: CO incurs larger label difficulty $l(K)$ but a proportionally smaller amplification, and the exponent of (16) rewards that trade. A practical workflow follows directly. Start from compliant-overdamped gains, monitor $\Gamma_T(K)\cdot\hat{L}_{\mathrm{va}}(K)$ rather than the training loss alone, and resolve the SO versus CU choice empirically because the ordering is fundamentally system-dependent.

B. Relation to Classical Analysis

Classical compounding-error analysis [8] bounds the performance gap by $O(\varepsilon T^2)$. Our amplification index refines the T^2 factor. For Schur-stable systems, $\Gamma_T(K) = O(1/(1-\rho(A_K))^2)$ remains bounded as $T \to \infty$ and converges to the squared $\mathcal{H}_2$ norm $\lim_{T\to\infty}\Gamma_T(K) = \|\mathcal{G}_K\|^2_{\mathcal{H}_2}$ [11], connecting the analysis to classical robust control. Block et al. [7] quantify how training noise compounds during autoregressive rollout, while our amplification index complements that line by exposing how the controller gains modulate the same pathway through A_K and B_K. The indeterminacy of the SO versus CU comparison matches the system-dependent observations reported in [1].

C. Extensions and Limitations

The linearization about the expert trajectory is the main simplification. Nonlinear, contact-rich settings are a natural next step via contraction theory [12]. The independence assumption on ξ_t weakens once covariate shift accumulates, and in that regime the bounds remain valid after replacing Σ^{roll}_K by an upper bound on the conditional sub-Gaussian proxy. Multi-step action chunks [2, 4] enter the framework unchanged by treating a length-H chunk as a single Hm-dimensional prediction error and evaluating $\Gamma_T(K)$ at the chunk-execution rate. For multi-joint systems with non-uniform gains the matrix-level comparison through $X_\infty(K)$ replaces the scalar reduction whenever the shape-preserving structure of Definition 4 fails to hold uniformly across joints.

VII. Conclusion

We have presented a theoretical framework that quantifies how controller gains shape behavior cloning failure through four results: sub-Gaussian error propagation, a closed-loop failure bound that factorizes into a gain-dependent amplification index and the validation loss, a scalar ordering reduction via shape-preserving upper-bound structure, and a global monotonicity theorem for canonical second-order systems covering both underdamped and overdamped regimes. The closed-form continuous-time stationary variance $X^{\mathrm{c}}_{\infty} = \sigma^2 K_p/(2K_d)$ recovers the stationary-variance result of [1] and extends the associated attenuation mechanism to nonasymptotic finite-horizon failure bounds. Natural extensions include nonlinear contact dynamics, co-optimization of gains with policy architecture, and tightening the discrete-time inheritance bound beyond the leading-order continuous-time limit.

Appendix I
Asymptotic Form of Γ_T for the Canonical System

For the canonical scalar second-order system under exact ZOH discretization with sampling period Δt, the closed-loop matrix A_K has eigenvalues

$$\lambda_{1,2}(K) = \exp\left(\frac{-\beta \pm \sqrt{\beta^2 - 4m\alpha}}{2m}\,\Delta t\right), \qquad (26)$$

which lie strictly inside the unit disk on the open Hurwitz region $\alpha, \beta > 0$. Bounding the geometric series of operator norms in (9) yields

$$\Gamma_T(K) \le \frac{\|B_K\|^2_{\mathrm{op}}}{1-\rho(A_K)^2}\Big(1-\rho(A_K)^{2T}\Big), \qquad (27)$$

where $\|B_K\|^2_{\mathrm{op}} = O(\Delta t^2)$ and $\rho(A_K)^2 = 1 - \beta\Delta t/m + O(\Delta t^2)$ as $\Delta t \to 0^+$. Substituting these expansions into (27) and using $1-\rho^{2T} \to 1$ in the long-horizon limit gives $\Gamma_T(K) \sim \alpha\Delta t/(2\beta)$, which reproduces the leading-order failure exponent of Corollary 11 and identifies X^{c}_{∞} as the rate constant of the long-horizon failure tail.

Appendix II
Multi-Joint Reduction via Definition 4

For an n-joint robot with diagonal mass $M = \mathrm{diag}(m_i)$ and diagonal gains $K_p = \mathrm{diag}(\alpha_i)$, $K_d = \mathrm{diag}(\beta_i)$, the closed-loop dynamics decouple per joint. Choosing $\bar{W} := I_{2n}$ and $\bar{\Sigma} := I_n$, the conditions of Definition 4 hold with

$$
\begin{aligned}
l(K) &= \lambda_{\max}(\Sigma_K^{\mathrm{roll}}), \\
b(K) &= \max_i \frac{\alpha_i^2\,\Delta t^2}{4m_i^2}, \\
\rho_*(K) &= \max_i \, e^{-\beta_i \Delta t/(2m_i)}.
\end{aligned}
$$

The scalar reduction of Theorem 5 therefore ranks multi-joint gains via the joint with the worst (largest) per-joint Ψ, recovering the canonical scalar comparison in the joint-decoupled case and providing a conservative ranking when cross-joint coupling is present in Σ_K^{roll}.

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